

DEPLOY AND OPERATE ODOO IN PRODUCTION

Infrastructure & DevOps

Docker, VPS, Nginx, SSL, and workers

Merit Advisory

Odoo Full-Stack Developer Program

Beginner-friendly lecture materials

July 2026

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Agenda

- 1 Section 1: Docker Fundamentals — 11 topics
- 2 Section 2: VPS and Reverse Proxy — 9 topics
- 3 Section 3: Odoo Performance Settings — 12 topics



1 Section 1

Docker Fundamentals

1.1 Containers vs VMs

- A container packages an app process and its dependencies.
- A VM packages a full guest operating system.
- Containers are lighter, but the host still matters.

Container mental model

app process + image filesystem + mounted volumes

VM mental model

guest OS + virtual disk + virtual network + app

1.2 Install Docker

- Install Docker from trusted package sources.
- Verify both the engine and compose plugin.
- Only trusted administrators should use the docker group.

Version checks

```
docker --version  
docker compose version
```

Service check

```
sudo systemctl enable --now docker  
sudo systemctl status docker
```

1.3 docker run, ps, and logs

- docker run starts a container from an image.
- docker ps shows running containers.
- docker logs is the first place to inspect startup errors.

Run a test container

```
docker run --rm hello-world
```

Inspect containers

```
docker ps  
docker logs odoo-web
```

1.4 Images vs Containers

- An image is the reusable blueprint.
- A container is a running or stopped instance.
- Removing a container should not remove persistent data.

List images

```
docker images
```

List all containers

```
docker ps -a
```

1.5 Dockerfile Odoo Idea

- A Dockerfile documents how to build a custom Odoo image.
- Install Python and system dependencies in the image.
- Keep addons and configuration separate from database data.

Minimal base

```
FROM odoo:18
USER root
RUN pip3 install --break-system-packages openpyxl
USER odoo
```

Copy addons

```
COPY ./custom_addons /mnt/extra-addons
```


1.6 Volumes

- Volumes keep data outside disposable containers.
- Odoo needs persistence for filestore and PostgreSQL data.
- Back up volumes before upgrades.

Named volume

```
volumes:  
  odoo-data:  
  pg-data:
```

Mount volume

```
services:  
  web:  
    volumes:  
      - odoo-data:/var/lib/odoo
```

1.7 Docker Network

- Docker networks let containers reach each other by service name.
- Keep databases off the public internet.
- Expose only the ports users or proxies need.

Compose network

```
networks:  
  odoo-net:
```

Database host name

```
db_host = db  
db_port = 5432
```

1.8 Compose File

- Compose describes multiple services in one YAML file.
- It is useful for Odoo, PostgreSQL, and helper services.
- Commit safe defaults, not secrets.

Compose command

```
docker compose up -d  
docker compose ps
```

Service skeleton

```
services:  
  web:  
    image: odoo:18  
  db:  
    image: postgres:16
```

1.9 Odoo and PostgreSQL Compose

- Odoo needs a PostgreSQL database service.
- Use environment variables for database connection settings.
- Persist both Odoo and PostgreSQL data.

Odoo service

```
web:  
  image: odoo:18  
  depends_on:  
    - db  
  ports:  
    - "8069:8069"
```

PostgreSQL service

```
db:  
  image: postgres:16  
  environment:  
    POSTGRES_DB: postgres  
    POSTGRES_USER: odoo
```

1.10 .env File

- .env keeps environment-specific values out of compose YAML.
- Do not commit real production secrets.
- Use clear variable names.

Example .env

```
ODOO_PORT=8069
POSTGRES_USER=odoo
POSTGRES_PASSWORD=change_me
```

Compose reference

```
ports:
  - "${ODOO_PORT}:8069"
environment:
  POSTGRES_PASSWORD: ${POSTGRES_PASSWORD}
```

1.11 Upgrade Containers Safely

- Back up the database and filestore before upgrading.
- Pull new images in staging first.
- Keep a rollback path for the previous version.

Pull and restart

```
docker compose pull  
docker compose up -d
```

Backup first

```
pg_dump -Fc odoo_prod > odoo_prod.dump  
tar -czf filestore.tgz /var/lib/odoo/filestore
```

2 Section 2

VPS and Reverse Proxy

2.1 Choose VPS Specs

- Size the VPS for users, workers, database, and backups.
- Prefer SSD storage for PostgreSQL.
- Leave room for growth and maintenance tasks.

Small training host

2 vCPU
4 GB RAM
60 GB SSD

Busier host

4 to 8 vCPU
16 GB RAM
separate backup storage

2.2 SSH Hardening

- Use SSH keys instead of passwords.
- Disable direct root login when possible.
- Keep a tested recovery path before changing access.

sshd settings

```
PasswordAuthentication no  
PermitRootLogin no
```

Reload SSH

```
sudo sshd -t  
sudo systemctl reload ssh
```

2.3 /opt/odoo Layout

- A predictable directory layout helps operations.
- Separate config, addons, backups, and logs.
- Document ownership and permissions.

Common layout

```
/opt/odoo/  
  config/  
  custom_addons/  
  backups/  
  logs/
```

Ownership idea

```
sudo chown -R odoo:odoo /opt/odoo
```

2.4 Nginx Reverse Proxy

- Nginx receives public HTTP traffic.
- It forwards requests to Odoo on the private port.
- It also handles TLS, headers, and upload limits.

Server block

```
server {  
    server_name odoo.example.com;  
    location / {  
        proxy_pass http://127.0.0.1:8069;  
    }  
}
```

Test config

```
sudo nginx -t  
sudo systemctl reload nginx
```

2.5 Proxy to 8069

- Port 8069 is the normal Odoo HTTP port.
- Expose it only to Nginx on production hosts.
- Set proxy_mode in Odoo when using a proxy.

Nginx proxy

```
location / {  
    proxy_set_header Host $host;  
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;  
    proxy_pass http://127.0.0.1:8069;  
}
```

Odoo config

```
proxy_mode = True  
xmlrpc_port = 8069
```

2.6 Longpolling 8072

- Odoo uses a separate longpolling or websocket endpoint. **Nginx longpolling**
- Route the longpolling path to port 8072 when configured.
- Chat and live updates depend on this path.

```
location /longpolling {  
    proxy_pass http://127.0.0.1:8072;  
}
```

Odoo port

```
longpolling_port = 8072
```

2.7 Certbot SSL

- Certbot can issue Let's Encrypt certificates.
- Use HTTPS for every production Odoo site.
- Verify automatic renewal.

Issue certificate

```
sudo certbot --nginx -d odoo.example.com
```

Renewal check

```
sudo certbot renew --dry-run
```

2.8 systemd odoo.service

- systemd keeps an Odoo process running on a VPS.
- It defines the user, command, and restart behavior.
- Logs can be viewed with journalctl.

Service command

```
[Service]
User=odoo
ExecStart=/opt/odoo/venv/bin/python /opt/odoo/odoo-bin -c
Restart=always
```

Operate service

```
sudo systemctl enable --now odoo
journalctl -u odoo -f
```

2.9 Cron Backup

- Backups should run automatically.
- Include both database dump and filestore.
- Test restores on another host.

Cron line

```
0 2 * * * /opt/odoo/scripts/backup.sh
```

Backup script idea

```
pg_dump -Fc odoo_prod > /backup/odoo_prod.dump  
tar -czf /backup/filestore.tgz /var/lib/odoo/filestore
```


3 Section 3

Odoo Performance Settings

3.1 Workers Idea

- Workers let Odoo handle multiple requests in parallel.
- Production deployments usually use workers.
- Cron workers and HTTP workers need enough memory.

Worker setting

```
workers = 5  
max_cron_threads = 1
```

No workers

```
workers = 0  
# common in simple development setups
```

3.2 Calculate Workers

- A common starting formula is CPU cores * 2 + 1.
- Adjust for memory, long requests, and cron load.
- Measure real traffic before increasing blindly.

Formula

```
workers = (cpu_cores * 2) + 1
```

Example

```
2 CPU cores -> 5 workers  
4 CPU cores -> 9 workers
```

3.3 Memory Limits

- Memory limits restart workers that grow too large.
- Set limits based on available RAM and worker count.
- Leave memory for PostgreSQL and the operating system.

Odoo limits

```
limit_memory_soft = 2147483648  
limit_memory_hard = 2684354560
```

Container limit

```
deploy:  
  resources:  
    limits:  
      memory: 4G
```

3.4 Time Limits

- Time limits stop requests that run too long.
- They protect workers from being stuck forever.
- Long imports may need a planned temporary setting.

Config values

```
limit_time_cpu = 60  
limit_time_real = 120
```

Import window idea

```
# raise limits only during a controlled import  
limit_time_real = 600
```

3.5 max_connections

- PostgreSQL max_connections must cover Odoo workers and tools.
- Too many connections can waste memory.
- Use pooling or careful sizing for larger systems.

PostgreSQL setting

```
max_connections = 200
```

Odoo db setting

```
db_maxconn = 64
```

3.6 Missing Indexes

- Missing indexes make common searches slow.
- Look for slow queries before adding indexes.
- Index fields used often in domains, joins, and sorting.

SQL index

```
CREATE INDEX training_course_state_idx  
ON training_course (state);
```

Odoo field index

```
state = fields.Selection(  
    selection=STATES,  
    index=True,  
)
```

3.7 pg_stat_statements

- pg_stat_statements records query statistics.
- It helps find slow and frequent SQL.
- Enable it before a performance incident if possible.

Enable extension

```
CREATE EXTENSION IF NOT EXISTS pg_stat_statements;
```

Find slow queries

```
SELECT calls, mean_exec_time, query  
FROM pg_stat_statements  
ORDER BY mean_exec_time DESC  
LIMIT 10;
```


3.8 Redis Sessions Idea

- Redis can store shared session or cache data in some designs.
- It is useful when multiple Odoo instances need shared state.
- Treat Redis as infrastructure that needs monitoring too.

Redis service

```
redis:  
  image: redis:7  
  restart: unless-stopped
```

Session concept

```
browser session -> load balancer -> Odoo nodes -> shared R
```

3.9 Load Balancing Idea

- Load balancing spreads traffic across Odoo nodes.
- All nodes must share database and filestore strategy.
- Start with one reliable node before adding more.

Upstream block

```
upstream odoo_backend {  
    server 10.0.0.11:8069;  
    server 10.0.0.12:8069;  
}
```

Proxy to upstream

```
location / {  
    proxy_pass http://odoo_backend;  
}
```

3.10 Monitoring Overview

- Monitoring shows whether the system is healthy.
- Track application, database, disk, and host metrics.
- Alerts should point to an action a person can take.

Useful signals

CPU, RAM, disk, HTTP errors, slow queries, backup age

Basic command checks

```
df -h  
free -m  
systemctl status odoo
```

3.11 Logging Overview

- Logs explain what happened before and during failures.
- Keep Odoo, Nginx, and PostgreSQL logs accessible.
- Rotate logs so disks do not fill.

Odoo log config

```
logfile = /var/log/odoo/odoo.log  
log_level = info
```

Journal logs

```
journalctl -u odoo --since "1 hour ago"
```

3.12 Restore Testing

- A backup is only useful if it can be restored.
- Restore tests prove database and filestore backups match.
- Schedule restore practice before emergencies.

Restore database

```
createdb odoo_restore  
pg_restore -d odoo_restore odoo_prod.dump
```

Restore filestore

```
mkdir -p /var/lib/odoo/filestore/odoo_restore  
tar -xzf filestore.tgz -C /var/lib/odoo/filestore/odoo_restore
```

NEXT STEP

Try the examples on your machine

Then open the next module deck

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07 · Infrastructure

